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VACUUM ADIABATIC BODY AND REFRIGERATOR

Abstract

A vacuum adiabatic body and a refrigerator are provided. The vacuum adiabatic body includes a support. The support includes a main support having a two-dimensional planar structure and crossing a space, and a first bar and a second bar, which respectively extend from both sides of the main support toward a first plate and a second plate, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation Application of prior U.S. patent application Ser. No. 18/124,038, filed Mar. 21, 2023, which is a continuation of application Ser. No. 16/635,758 filed Jan. 31, 2020, which is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2018/008691, filed Jul. 31, 2018, which claims priority to Korean Patent Application No. 10-2017-0097821, filed Aug. 1, 2017, whose entire disclosures are hereby incorporated by reference.

BACKGROUND

Field

[0002] A vacuum adiabatic body and a refrigerator are disclosed herein.

Background

[0003] A vacuum adiabatic body is a product for suppressing heat transfer by vacuumizing an interior of a body thereof. The vacuum adiabatic body may reduce heat transfer by convection and conduction, and hence is applied to heating apparatuses and refrigerating apparatuses. In a typical adiabatic method applied to a refrigerator, although it is differently applied in refrigeration and freezing, a foam urethane adiabatic wall having a thickness of about 30 cm or more is generally provided. However, an internal volume of the refrigerator is therefore reduced.

[0004] In order to increase the internal volume of a refrigerator, there is an attempt to apply a vacuum adiabatic body to the refrigerator.

[0005] First, Korean Patent No. 10-0343719 (Reference Document 1) of the present applicant has been disclosed. According to Reference Document 1, there is disclosed a method in which a vacuum adiabatic panel is prepared and then built in walls of a refrigerator, and an exterior of the vacuum adiabatic panel is finished with a separate molding as Styrofoam. According to the method, additional foaming is not required, and adiabatic performance of the refrigerator is improved. However, fabrication cost is increased, and a fabrication method is complicated. As another example, a technique of providing walls using a vacuum adiabatic material and additionally providing adiabatic walls using a foam filling material has been disclosed in Korean Patent Publication No. 10-2015-0012712 (Reference Document 2). According to Reference Document 2, fabrication cost is increased, and a fabrication method is complicated.

[0006] As further another example, there is an attempt to fabricate all walls of a refrigerator using a vacuum adiabatic body that is a single product. For example, a technique of providing an adiabatic structure of a refrigerator to be in a vacuum state has been disclosed in U.S. Patent Laid-Open Publication No. US2040226956A1 (Reference Document 3). However, it is difficult to obtain a practical level of an adiabatic effect by providing a wall of the refrigerator with sufficient vacuum. In detail, there are limitations in that it is difficult to prevent a heat transfer phenomenon at a contact portion between an outer case and an inner case having different temperatures, it is difficult to maintain a stable vacuum state, and it is difficult to prevent deformation of a case due to a negative pressure of the vacuum state. Due to these limitations, the technology disclosed in Reference Document 3 is limited to a cryogenic refrigerator, and does not provide a level of technology applicable to general households.

[0007] The present applicant had filed Patent Application No. 10-2011-0113414 (Reference Document 4) in consideration of the above-described limitations. Reference Document 4 proposes a refrigerator including a vacuum adiabatic body. Particularly, a space maintenance member for installing a radiation resistance sheet is built.

[0008] According to the document, it is difficult to install the radiation resistance sheet in a

supporting unit, particularly, when the radiation resistance sheet is inserted, the space maintenance member for maintaining a space has to be separately inserted. In addition, as a member made of a resin material is used, a weight, cost, and outgassing increase. Further, as the space maintenance member having a predetermined thickness has to be installed, there is a limitation in securing an adiabatic thickness of the vacuum adiabatic body.

[0009] Embodiments provide a vacuum adiabatic body, which is convenient in installation of a radiation resistance sheet in a supporting unit, and a refrigerator. Embodiments also provide a vacuum adiabatic body, in which a plurality of radiation resistance sheets is conveniently coupled, and a refrigerator. Embodiments provide a vacuum adiabatic body, in which a possibility of damage is reduced, and a refrigerator. Embodiments also provide a vacuum adiabatic body, which is improved in productivity when being assembled, and a refrigerator.

[0010] In order to conveniently install a radiation resistance sheet to a supporting unit, the supporting unit may include two support plates, and the radiation resistance sheet may be supported by a support protrusion provided on a bar coupling the two support plates to each other and thus be maintained in position in an interval part between the plate members. In order to conveniently couple a plurality of radiation resistance sheets, at least two kinds of bars may be provided in the supporting unit. In order to prevent the supporting unit from being damaged, an A type bar provided by coupling a high column provided on the support plate of one side to a support groove provided in the support plate of the other side and a B type bar provided by coupling a multiple high column having a large diameter portion and a small diameter portion, which have diameters different from each other, to an accommodation part provided on the support plate of the other side to support the small diameter portion may be provided.

[0011] The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

[0012] According to embodiments, a worker may conveniently assemble the supporting unit and the radiation resistance sheet together with each other. According to embodiments, damage of parts may be reduced to improve product yield. According to embodiments, parts may be shared to reduce stock cost.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a perspective view of a refrigerator according to an embodiment;

[0014] FIG. 2 is a view schematically showing a vacuum adiabatic body used in a main body and a door of the refrigerator;

[0015] FIGS. 3A-3B are views illustrating various embodiments of an internal configuration of a vacuum space part;

[0016] FIG. 4 is a diagram illustrating results obtained by examining resins;

[0017] FIG. 5 illustrates results obtained by performing an experiment on vacuum maintenance performances of resins;

[0018] FIGS. 6A-6C are results obtained by analyzing components of gases discharged from polyphenylene sulfide (PPS) and low outgassing polycarbonate (PC);

[0019] FIG. 7 illustrates results obtained by measuring maximum deformation temperatures at which resins are damaged by atmospheric pressure in high-temperature exhaustion;

[0020] FIGS. 8A-8C are views showing various embodiments of conductive resistance sheets and peripheral parts thereof;

[0021] FIG. 9 is a cross-sectional view of a supporting unit according to an embodiment.

[0022] FIG. 10 is a view for explaining an A type bar.

[0023] FIG. **11** is a view for explaining a B-11 type bar.

[0024] FIG. **12** is a view for explaining a B-12 type bar.

[0025] FIG. **13** is a view for explaining a coupling process of the supporting unit.

[0026] FIG. **14** is a cross-sectional view of a supporting unit according to another embodiment.

[0027] FIG. **15** is a cross-sectional view of a supporting unit according to further another embodiment.

[0028] FIG. **16** is a view for explaining a C type bar.

[0029] FIG. **17** is a view for explaining a different C type bar.

[0030] FIG. **18** illustrates graphs showing changes in adiabatic performance and changes in gas conductivity with respect to vacuum pressures by applying a simulation.

[0031] FIG. **19** is a graph illustrating results obtained by observing a time and a pressure in a process of exhausting an inside of the vacuum adiabatic body when a supporting unit is used.

[0032] FIG. **20** is a graph illustrating results obtained by comparing a vacuum pressure with gas conductivity.

DETAILED DESCRIPTION

[0033] Hereinafter, exemplary embodiments will be described with reference to the accompanying drawings. The embodiments may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein, and a person of ordinary skill in the art, who understands the spirit, may readily implement other embodiments included within the scope of the same concept by adding, changing, deleting, and adding components; rather, it will be understood that they are also included within the scope.

[0034] The drawings shown below may be displayed differently from the actual product, or exaggerated or simple or detailed parts may be deleted, but this is intended to facilitate understanding of the technical idea. It should not be construed as limited.

[0035] In the following description, the term vacuum pressure means any pressure state lower than atmospheric pressure. In addition, the expression that a vacuum degree of A is higher than that of B means that a vacuum pressure of A is lower than that of B.

[0036] FIG. **1** is a perspective view of a refrigerator according to an embodiment. Referring to FIG. **1**, the refrigerator **1** includes a main body **2** provided with a cavity **9** capable of storing storage goods and a door **3** provided to open/close the main body **2**. The door **3** may be rotatably or slidably movably disposed to open/close the cavity **9**. The cavity **9** may provide at least one of a refrigerating compartment or a freezing compartment.

[0037] Parts constituting a freezing cycle in which cold air is supplied into the cavity **9**. In detail, the parts include a compressor **4** that compresses a refrigerant, a condenser **5** that condenses the compressed refrigerant, an expander **6** that expands the condensed refrigerant, and an evaporator **7** that evaporates the expanded refrigerant to take heat. As a typical structure, a fan may be installed at a position adjacent to the evaporator **7**, and a fluid blown from the fan may pass through the evaporator **7** and then be blown into the cavity **9**. A freezing load is controlled by adjusting a blowing amount and blowing direction by the fan, adjusting an amount of a circulated refrigerant, or adjusting a compression rate of the compressor, so that it is possible to control a refrigerating space or a freezing space.

[0038] FIG. **2** is a view schematically showing a vacuum adiabatic body used in the main body and the door of the refrigerator. In FIG. **2**, a main body-side vacuum adiabatic body is illustrated in a state in which top and side walls are removed, and a door-side vacuum adiabatic body is illustrated in a state in which a portion of a front wall is removed. In addition, sections of portions at conductive resistance sheets are schematically illustrated for convenience of understanding.

[0039] Referring to FIG. **2**, the vacuum adiabatic body includes a first plate member (first plate) **10** for providing a wall of a low-temperature space, a second plate member (second plate) **20** for providing a wall of a high-temperature space, a vacuum space part (vacuum space) **50** defined as an interval part between the first and second plate members **10** and **20**. Also, the vacuum adiabatic

body includes conductive resistance sheets **60** and **63** for preventing heat conduction between the first and second plate members **10** and **20**. A sealing part (sealing) **61** for sealing the first and second plate members **10** and **20** is provided such that the vacuum space part **50** is in a sealed state. When the vacuum adiabatic body is applied to a refrigerating or heating cabinet, the first plate member **10** may be referred to as an inner case, and the second plate member **20** may be referred to as an outer case. A machine room **8** in which parts providing a freezing cycle are accommodated is placed at a lower rear side of the main body-side vacuum adiabatic body, and an exhaust port **40** for forming a vacuum state by exhausting air in the vacuum space part **50** is provided at any one side of the vacuum adiabatic body. In addition, a pipeline **64** passing through the vacuum space part **50** may be further installed so as to install a defrosting water line and electric lines.

[0040] The first plate member **10** may define at least one portion of a wall for a first space provided thereto. The second plate member **20** may define at least one portion of a wall for a second space provided thereto. The first space and the second space may be defined as spaces having different temperatures. The wall for each space may serve as not only a wall directly contacting (facing) the space but also a wall not contacting (facing) the space. For example, the vacuum adiabatic body of the embodiment may also be applied to a product further having a separate wall contacting (facing) each space.

[0041] Factors of heat transfer, which cause loss of the adiabatic effect of the vacuum adiabatic body, are heat conduction between the first and second plate members **10** and **20**, heat radiation between the first and second plate members **10** and **20**, and gas conduction of the vacuum space part **50**. Hereinafter, a heat resistance unit provided to reduce adiabatic loss related to the factors of the heat transfer will be provided. The vacuum adiabatic body and the refrigerator of the embodiment do not exclude that another adiabatic means is further provided to at least one side of the vacuum adiabatic body. Therefore, an adiabatic means using foaming, for example, may be further provided to another side of the vacuum adiabatic body.

[0042] FIGS. **3A-32B** are views illustrating various embodiments of an internal configuration of the vacuum space part. Referring to FIG. **3A**, the vacuum space part **50** may be provided in a third space having a pressure different from that of each of the first and second spaces, for example, a vacuum state, thereby reducing adiabatic loss. The third space may be provided at a temperature between a temperature of the first space and a temperature of the second space. As the third space is provided as a space in the vacuum state, the first and second plate members **10** and **20** receive a force contracting in a direction in which they approach each other due to a force corresponding to a pressure difference between the first and second spaces. Therefore, the vacuum space part **50** may be deformed in a direction in which it is reduced. In this case, the adiabatic loss may be caused due to an increase in amount of heat radiation, caused by the contraction of the vacuum space part **50**, and an increase in amount of heat conduction, caused by contact between the plate members **10** and **20**.

[0043] A supporting unit (support) **30** may be provided to reduce deformation of the vacuum space part **50**. The supporting unit **30** includes a bar **31**. The bar **31** may extend in a substantially vertical direction with respect to the plate members to support a distance between the first plate member and the second plate member. A support plate **35** may be additionally provided on at least any one end of the bar **31**. The support plate **35** may connect at least two or more bars **31** to each other to extend in a horizontal direction with respect to the first and second plate members **10** and **20**. The support plate **35** may be provided in a plate shape or may be provided in a lattice shape so that an area of the support plate contacting the first or second plate member **10** or **20** decreases, thereby reducing heat transfer. The bars **31** and the support plate **35** are fixed to each other at at least one portion, to be inserted together between the first and second plate members **10** and **20**. The support plate **35** contacts at least one of the first and second plate members **10** and **20**, thereby preventing deformation of the first and second plate members **10** and **20**. In addition, based on an extending direction of the bars **31**, a total sectional area of the support plate **35** is provided to be greater than

that of the bars **31**, so that heat transferred through the bars **31** may be diffused through the support plate **35**.

[0044] A material of the supporting unit **30** will be described hereinafter.

[0045] The supporting unit **30** is to have a high compressive strength so as to endure the vacuum pressure. Also, the supporting unit **30** is to have a low outgassing rate and a low water absorption rate so as to maintain the vacuum state. Further, the supporting unit **30** is to have a low thermal conductivity so as to reduce heat conduction between the plate members. Furthermore, the supporting unit **30** is to secure the compressive strength at a high temperature so as to endure a high-temperature exhaust process. Additionally, the supporting unit **30** is to have an excellent machinability so as to be subjected to molding. Also, the supporting unit **30** is to have a low cost for molding. A time required to perform the exhaust process takes about a few days. Hence, the time is reduced, thereby considerably improving fabrication cost and productivity. Therefore, the compressive strength is to be secured at the high temperature because an exhaust speed is increased as a temperature at which the exhaust process is performed becomes higher. The inventor has performed various examinations under the above-described conditions.

[0046] First, ceramic or glass has a low outgassing rate and a low water absorption rate, but its machinability is remarkably lowered. Hence, the ceramic and glass may not be used as the material of the supporting unit **30**. Therefore, resin may be considered as the material of the supporting unit **30**.

[0047] FIG. **4** is a diagram illustrating results obtained by examining resins. Referring to FIG. **4**, the present inventor has examined various resins, and most of the resins cannot be used because their outgassing rates and water absorption rates are remarkably high. Accordingly, the present inventor has examined resins that approximately satisfy conditions of the outgassing rate and the water absorption rate. As a result, polyethylene resin (PE) is inappropriate to be used due to its high outgassing rate and its low compressive strength. Polychlorotrifluoroethylene (PCTFE) is not used due to its remarkably high price. Polyether ether ketone (PEEK) is inappropriate to be used due to its high outgassing rate. Accordingly, it is determined that that a resin selected from the group consisting of polycarbonate (PC), glass fiber PC, low outgassing PC, polyphenylene sulfide (PPS), and liquid crystal polymer (LCP) may be used as the material of the supporting unit. However, an outgassing rate of the PC is 0.19, which is at a low level. Hence, as the time required to perform baking in which exhaustion is performed by applying heat is increased to a certain level, the PC may be used as the material of the supporting unit.

[0048] The present inventor has found an optimal material by performing various studies on resins expected to be used inside the vacuum space part. Hereinafter, results of the performed studies will be described with reference to the accompanying drawings.

[0049] FIG. **5** is a view illustrating results obtained by performing an experiment on vacuum maintenance performances of the resins. Referring to FIG. **5**, there is illustrated a graph showing results obtained by fabricating the supporting unit using the respective resins and then testing vacuum maintenance performances of the resins. First, a supporting unit fabricated using a selected material was cleaned using ethanol, left at a low pressure for 48 hours, exposed to air for 2.5 hours, and then subjected to an exhaust process at 90° C. for about 50 hours in a state that the supporting unit was put in the vacuum adiabatic body, thereby measuring a vacuum maintenance performance of the supporting unit.

[0050] It may be seen that in the case of the LCP, its initial exhaust performance is best, but its vacuum maintenance performance is bad. It may be expected that this is caused by sensitivity of the LCP to temperature. Also, it is expected through characteristics of the graph that, when a final allowable pressure is 5×10^{-3} Torr, its vacuum performance will be maintained for a time of about 0.5 year. Therefore, the LCP is inappropriate as the material of the supporting unit.

[0051] It may be seen that, in the case of the glass fiber PC (G/F PC), its exhaust speed is fast, but its vacuum maintenance performance is low. It is determined that this will be influenced by an

additive. Also, it is expected through the characteristics of the graph that the glass fiber PC will maintain its vacuum performance will be maintained under the same condition for a time of about 8.2 years. Therefore, the LCP is inappropriate as the material of the supporting unit.

[0052] It is expected that, in the case of the low outgassing PC (L/O PC), its vacuum maintenance performance is excellent, and its vacuum performance will be maintained under the same condition for a time of about 34 years, as compared with the above-described two materials. However, it may be seen that the initial exhaust performance of the low outgassing PC is low, and therefore, fabrication efficiency of the low outgassing PC is lowered.

[0053] It may be seen that, in the case of the PPS, its vacuum maintenance performance is remarkably excellent, and its exhaust performance is also excellent. Therefore, it is considered that, based on the vacuum maintenance performance, the PPS is used as the material of the supporting unit.

[0054] FIGS. 6A-6C illustrate results obtained by analyzing components of gases discharged from the PPS and the low outgassing PC, in which the horizontal axis represents mass numbers of gases and the vertical axis represents concentrations of gases. FIG. 6A illustrates a result obtained by analyzing a gas discharged from the low outgassing PC. In FIG. 6A, it may be seen that H.sub.2 series (I), H.sub.2O series (II), N.sub.2/CO/CO.sub.2/O.sub.2 series (III), and hydrocarbon series (IV) are equally discharged. FIG. 6B illustrates a result obtained by analyzing a gas discharged from the PPS. In FIG. 6B, it may be seen that H.sub.2 series (I), H.sub.2O series (II), and N.sub.2/CO/CO.sub.2/O.sub.2 series (III) are discharged to a weak extent. FIG. 6C is a result obtained by analyzing a gas discharged from stainless steel. In FIG. 6C, it may be seen that a similar gas to the PPS is discharged from the stainless steel. Consequently, it may be seen that the PPS discharges a similar gas to the stainless steel. As the analyzed result, it may be re-confirmed that the PPS is excellent as the material of the supporting unit.

[0055] FIG. 7 illustrates results obtained by measuring maximum deformation temperatures at which resins are damaged by atmospheric pressure in high-temperature exhaustion. The bars 31 were provided at a diameter of 2 mm at a distance of 30 mm. Referring to FIG. 7, it may be seen that a rupture occurs at 60° C. in the case of the PE, a rupture occurs at 90° C. in the case of the low outgassing PC, and a rupture occurs at 125° C. in the case of the PPS. As the analyzed result, it may be seen that the PPS is most used as the resin used inside of the vacuum space part. However, the low outgassing PC may be used in terms of fabrication cost.

[0056] A radiation resistance sheet 32 for reducing heat radiation between the first and second plate members 10 and 20 through the vacuum space part 50 will be described. The first and second plate members 10 and 20 may be made of a stainless material capable of preventing corrosion and providing a sufficient strength. The stainless material has a relatively high emissivity of 0.16, and hence a large amount of radiation heat may be transferred. In addition, the supporting unit 30 made of the resin has a lower emissivity than the plate members, and is not entirely provided to inner surfaces of the first and second plate members 10 and 20. Hence, the supporting unit 30 does not have great influence on radiation heat. Therefore, the radiation resistance sheet 32 may be provided in a plate shape over a majority of the area of the vacuum space part 50 so as to concentrate on reduction of radiation heat transferred between the first and second plate members 10 and 20. A product having a low emissivity may be preferably used as the material of the radiation resistance sheet 32. In an embodiment, an aluminum foil having an emissivity of 0.02 may be used as the radiation resistance sheet 32. Also, as the transfer of radiation heat may not be sufficiently blocked using one radiation resistance sheet, at least two radiation resistance sheets 32 may be provided at a certain distance so as not to contact each other. Also, at least one radiation resistance sheet may be provided in a state in which it contacts the inner surface of the first or second plate member 10 or 20.

[0057] Referring back to FIG. 3B, a distance between the plate members is maintained by the supporting unit 30, and a porous material 33 may be filled in the vacuum space part 50. The porous

material **33** may have a higher emissivity than the stainless material of the first and second plate members **10** and **20**. However, as the porous material **33** is filled in the vacuum space part **50**, the porous material **33** has a high efficiency for resisting the radiation heat transfer. In the present embodiment, the vacuum adiabatic body may be manufactured without the radiation resistance sheet **32**.

[0058] FIGS. **8A-8C** are views showing various embodiments of conductive resistance sheets and peripheral parts thereof. Structures of the conductive resistance sheets are briefly illustrated in FIG. **2**, but will be understood in detail with reference to the drawings.

[0059] First, a conductive resistance sheet proposed in FIG. **8A** may be applied to the main body-side vacuum adiabatic body. Specifically, the first and second plate members **10** and **20** are to be sealed so as to vacuumize the interior of the vacuum adiabatic body. In this case, as the two plate members have different temperatures from each other, heat transfer may occur between the two plate members. Conductive resistance sheet **60** is provided to prevent heat conduction between two different kinds of plate members.

[0060] The conductive resistance sheet **60** may be provided with sealing parts **61** at which both ends of the conductive resistance sheet **60** are sealed to defining at least one portion of the wall for the third space and maintain the vacuum state. The conductive resistance sheet **60** may be provided as a thin foil in unit of micrometer so as to reduce the amount of heat conducted along the wall for the third space. The sealing parts **610** may be provided as welding parts. That is, the conductive resistance sheet **60** and the plate members **10** and **20** may be fused to each other. In order to cause a fusing action between the conductive resistance sheet **60** and the plate members **10** and **20**, the conductive resistance sheet **60** and the plate members **10** and **20** may be made of the same material, and a stainless material may be used as the material. The sealing parts **610** are not limited to the welding parts, and may be provided through a process, such as cocking. The conductive resistance sheet **60** may be provided in a curved shape. Thus, a heat conduction distance of the conductive resistance sheet **60** is provided longer than a linear distance of each plate member, so that the amount of heat conduction may be further reduced.

[0061] A change in temperature occurs along the conductive resistance sheet **60**. Therefore, in order to block heat transfer to the exterior of the conductive resistance sheet **60**, a shielding part (shield) **62** may be provided at an exterior of the conductive resistance sheet **60** such that an adiabatic action occurs. In other words, in the refrigerator, the second plate member **20** has a high temperature and the first plate member **10** has a low temperature. In addition, heat conduction from high temperature to low temperature occurs in the conductive resistance sheet **60**, and hence, the temperature of the conductive resistance sheet **60** is suddenly changed. Therefore, when the conductive resistance sheet **60** is opened to the exterior thereof, heat transfer through the opened place may seriously occur. In order to reduce heat loss, the shielding part **62** is provided at the exterior of the conductive resistance sheet **60**. For example, when the conductive resistance sheet **60** is exposed to any one of the low-temperature space and the high-temperature space, the conductive resistance sheet **60** does not serve as a conductive resistor as well as the exposed portion thereof.

[0062] The shielding part **62** may be provided as a porous material contacting an outer surface of the conductive resistance sheet **60**. The shielding part **62** may be provided as an adiabatic structure, e.g., a separate gasket, which is placed at the exterior of the conductive resistance sheet **60**. The shielding part **62** may be provided as a portion of the vacuum adiabatic body, which is provided at a position facing a corresponding conductive resistance sheet **60** when the main body-side vacuum adiabatic body is closed with respect to the door-side vacuum adiabatic body. In order to reduce heat loss even when the main body and the door are opened, the shielding part **62** may be provided as a porous material or a separate adiabatic structure.

[0063] A conductive resistance sheet proposed in FIG. **8B** may be applied to the door-side vacuum adiabatic body. In FIG. **8B**, portions different from those of FIG. **8A** are described, and the same

description is applied to portions identical to those of FIG. 8A. A side frame **70** is further provided at an outside of the conductive resistance sheet **60**. A part for sealing between the door and the main body, an exhaust port necessary for an exhaust process, a getter port for vacuum maintenance, for example, may be placed on the side frame **70**. This is because mounting of parts is convenient in the main body-side vacuum adiabatic body, but mounting positions of parts are limited in the door-side vacuum adiabatic body.

[0064] In the door-side vacuum adiabatic body, it is difficult to place the conductive resistance sheet **60** at a front end portion (front end) of the vacuum space part, i.e., a corner side portion (corner side) of the vacuum space part. This is because, unlike the main body, a corner edge portion (corner edge) of the door is exposed to the exterior. More specifically, if the conductive resistance sheet **60** is placed at the front end portion of the vacuum space part, the corner edge portion of the door is exposed to the exterior, and hence, there is a disadvantage in that a separate adiabatic part should be configured so as to heat-insulate the conductive resistance sheet **60**.

[0065] A conductive resistance sheet proposed in FIG. 8C may be installed in the pipeline passing through the vacuum space part. In FIG. 8C, portions different from those of FIGS. 8A and 8B are described, and the same description is applied to portions identical to those of FIGS. 8A and 8B. A conductive resistance sheet having the same shape as that of FIG. 8A, a wrinkled conductive resistance sheet **63** may be provided at a peripheral portion of the pipeline **64**. Accordingly, a heat transfer path may be lengthened, and deformation caused by a pressure difference may be prevented. In addition, a separate shielding part may be provided to improve adiabatic performance of the conductive resistance sheet.

[0066] A heat transfer path between the first and second plate members **10** and **20** will be described with reference back to FIG. 8A. Heat passing through the vacuum adiabatic body may be divided into surface conduction heat {circle around (1)} conducted along a surface of the vacuum adiabatic body, more specifically, the conductive resistance sheet **60**, supporter conduction heat {circle around (2)} conducted along the supporting unit **30** provided inside of the vacuum adiabatic body, gas conduction heat {circle around (3)} conducted through an internal gas in the vacuum space part, and radiation transfer heat {circle around (4)} transferred through the vacuum space part.

[0067] The transfer heat may be changed depending on various design dimensions. For example, the supporting unit may be changed such that the first and second plate members **10** and **20** may endure a vacuum pressure without being deformed, the vacuum pressure may be changed, a distance between the plate members may be changed, and a length of the conductive resistance sheet may be changed. The transfer heat may be changed depending on a difference in temperature between the spaces (the first and second spaces) respectively provided by the plate members. In the embodiment, a configuration of the vacuum adiabatic body has been found by considering that its total heat transfer amount is smaller than that of a typical adiabatic structure formed by foaming polyurethane. In a typical refrigerator including the adiabatic structure formed by foaming the polyurethane, an effective heat transfer coefficient may be proposed as 19.6 mW/mK.

[0068] By performing a relative analysis on heat transfer amounts of the vacuum adiabatic body of the embodiment, a heat transfer amount by the gas conduction heat {circle around (3)} may become smallest. For example, the heat transfer amount by the gas conduction heat {circle around (3)} may be controlled to be equal to or smaller than 4% of the total heat transfer amount. A heat transfer amount by solid conduction heat defined as a sum of the surface conduction heat {circle around (1)} and the supporter conduction heat {circle around (2)} is largest. For example, the heat transfer amount by the solid conduction heat may reach 75% of the total heat transfer amount. A heat transfer amount by the radiation transfer heat {circle around (4)} is smaller than the heat transfer amount by the solid conduction heat but larger than the heat transfer amount of the gas conduction heat. For example, the heat transfer amount by the radiation transfer heat {circle around (4)} may occupy about 20% of the total heat transfer amount.

[0069] According to such a heat transfer distribution, effective heat transfer coefficients (ek: effective K) (W/mK) of the surface conduction heat {circle around (1)}, the supporter conduction heat {circle around (2)}, the gas conduction heat {circle around (3)}, and the radiation transfer heat {circle around (4)} may have an order of Math Equation 1.

$$[00001] \ eK_{\text{solidconductionheat}} > eK_{\text{radiationtransferheat}} > eK_{\text{gasconductionheat}} \quad [\text{Equation1}]$$

[0070] Here, the effective heat transfer coefficient (ek) is a value that may be measured using a shape and temperature differences of a target product. The effective heat transfer coefficient (ek) is a value that may be obtained by measuring a total heat transfer amount and a temperature at least one portion at which heat is transferred. For example, a calorific value (W) is measured using a heating source that may be quantitatively measured in the refrigerator, a temperature distribution (K) of the door is measured using heats respectively transferred through a main body and an edge of the door of the refrigerator, and a path through which heat is transferred is calculated as a conversion value (m), thereby evaluating an effective heat transfer coefficient.

[0071] The effective heat transfer coefficient (ek) of the entire vacuum adiabatic body is a value given by $k=QL/A\Delta T$. Here, Q denotes a calorific value (W) and may be obtained using a calorific value of a heater. A denotes a sectional area (m²) of the vacuum adiabatic body, L denotes a thickness (m) of the vacuum adiabatic body, and ΔT denotes a temperature difference.

[0072] For the surface conduction heat, a conductive calorific value may be obtained through a temperature difference (ΔT) between an entrance and an exit of the conductive resistance sheet **60** or **63**, a sectional area (A) of the conductive resistance sheet, a length (L) of the conductive resistance sheet, and a thermal conductivity (k) of the conductive resistance sheet (the thermal conductivity of the conductive resistance sheet is a material property of a material and may be obtained in advance). For the supporter conduction heat, a conductive calorific value may be obtained through a temperature difference (ΔT) between an entrance and an exit of the supporting unit **30**, a sectional area (A) of the supporting unit, a length (L) of the supporting unit, and a thermal conductivity (k) of the supporting unit. Here, the thermal conductivity of the supporting unit is a material property of a material and may be obtained in advance. The sum of the gas conduction heat {circle around (3)}, and the radiation transfer heat {circle around (4)} may be obtained by subtracting the surface conduction heat and the supporter conduction heat from the heat transfer amount of the entire vacuum adiabatic body. A ratio of the gas conduction heat {circle around (3)}, and the radiation transfer heat {circle around (4)} may be obtained by evaluating radiation transfer heat when no gas conduction heat exists by remarkably lowering a vacuum degree of the vacuum space part **50**.

[0073] When a porous material is provided inside the vacuum space part **50**, porous material conduction heat {circle around (5)} may be a sum of the supporter conduction heat {circle around (2)} and the radiation transfer heat {circle around (4)}. The porous material conduction heat may be changed depending on various variables including a kind, and an amount, for example, of the porous material.

[0074] According to an embodiment, a temperature difference $\Delta T1$ between a geometric center formed by adjacent bars **31** and a point at which each of the bars **31** is located may be provided to be less than 0.5° C. Also, a temperature difference $\Delta T2$ between the geometric center formed by the adjacent bars **31** and an edge portion of the vacuum adiabatic body may be provided to be less than 0.5° C. In the second plate member **20**, a temperature difference between an average temperature of the second plate and a temperature at a point at which a heat transfer path passing through the conductive resistance sheet **60** or **63** meets the second plate may be largest. For example, when the second space is a region hotter than the first space, the temperature at the point at which the heat transfer path passing through the conductive resistance sheet meets the second plate member becomes lowest. Similarly, when the second space is a region colder than the first space, the temperature at the point at which the heat transfer path passing through the conductive resistance

sheet meets the second plate member becomes highest.

[0075] This means that the amount of heat transferred through other points except the surface conduction heat passing through the conductive resistance sheet should be controlled, and the entire heat transfer amount satisfying the vacuum adiabatic body may be achieved only when the surface conduction heat occupies the largest heat transfer amount. To this end, a temperature variation of the conductive resistance sheet may be controlled to be larger than that of the plate member.

[0076] Physical characteristics of the parts constituting the vacuum adiabatic body will be described. In the vacuum adiabatic body, a force by vacuum pressure is applied to all of the parts. Therefore, a material having a strength (N/m²) of a certain level may be used.

[0077] Under such circumstances, the plate members **10** and **20** and the side frame **70** may be made of a material having a sufficient strength with which they are not damaged by even vacuum pressure. For example, when the number of bars **31** is decreased so as to limit support conduction heat, deformation of the plate member occurs due to the vacuum pressure, which may negatively influence the external appearance of refrigerator. The radiation resistance sheet **32** may be made of a material that has a low emissivity and may be easily subjected to thin film processing. Also, the radiation resistance sheet **32** is to ensure a sufficient strength not to be deformed by an external impact. The supporting unit **30** is provided with a strength sufficient so as to support the force of the vacuum pressure and endure an external impact, and is to have machinability. The conductive resistance sheet **60** may be made of a material that has a thin plate shape and may endure the vacuum pressure.

[0078] In an embodiment, the plate member, the side frame, and the conductive resistance sheet may be made of stainless materials having the same strength. The radiation resistance sheet may be made of aluminum having a weaker strength than the stainless materials. The supporting unit may be made of resin having a weaker strength than the aluminum.

[0079] Unlike the strength from the point of view of materials, analysis from the point of view of stiffness is required. The stiffness (N/m) is a property that would not be easily deformed. Although the same material is used, its stiffness may be changed depending on its shape. The conductive resistance sheets **60** or **63** may be made of a material having a strength, but the stiffness of the material is low so as to increase heat resistance and minimize radiation heat as the conductive resistance sheet is uniformly spread without any roughness when the vacuum pressure is applied. The radiation resistance sheet **32** requires a stiffness of a certain level so as not to contact another part due to deformation. Particularly, an edge portion (edge) of the radiation resistance sheet may generate conduction heat due to drooping caused by self-load of the radiation resistance sheet. Therefore, a stiffness of a certain level is required. The supporting unit **30** requires a stiffness enough to endure a compressive stress from the plate member and an external impact.

[0080] In an embodiment, the plate member and the side frame may have the highest stiffness so as to prevent deformation caused by the vacuum pressure. The supporting unit, particularly, the bar may have the second highest stiffness. The radiation resistance sheet may have a stiffness that is lower than that of the supporting unit but higher than that of the conductive resistance sheet. Lastly, the conductive resistance sheet may be made of a material that is easily deformed by the vacuum pressure and has the lowest stiffness. Even when the porous material **33** is filled in the vacuum space part **50**, the conductive resistance sheet may have the lowest stiffness, and the plate member and the side frame may have the highest stiffness.

[0081] Hereinafter, a vacuum pressure determined depending on an internal state of the vacuum adiabatic body is discussed. As already described above, a vacuum pressure is to be maintained inside of the vacuum adiabatic body so as to reduce heat transfer. The vacuum pressure is maintained as low as possible so as to reduce the heat transfer.

[0082] The vacuum space part may resist heat transfer used only the supporting unit **30**. A porous material **33** may be filled within the supporting unit inside of the vacuum space part **50** to resist to heat transfer. Heat transfer to the porous material may be resisted without applying the supporting

unit.

[0083] In the above description, as a material suitable for the supporting unit, a resin of PPS has been proposed. These resins often have poor fluidity of the resin during molding, so that a shape of the molded article often fails to properly obtain a desired shape. The bar **31** is provided on the support plate **35** at intervals of 2 cm to 3 cm, and the bar **31** has a height of 1 cm to 2 cm.

[0084] The bar is inserted into the vacuum adiabatic body in a state in which the bar is not completely coupled to the support plate due to the material and characteristics of the construction described above. Thereafter, when being exhausted, the support plate and the bar are completely coupled to each other due to a vacuum pressure. However, there is a fear that damage may occur due to warpage of the bar during the coupling process.

[0085] In order to fix the height of the radiation resistance sheet **32**, an interval maintenance member is inserted into a gap between the radiation resistance sheets. If the interval maintenance member is provided as a separate member and inserted into the bar **31**, convenience of assembly may deteriorate.

[0086] Hereinafter, a supporting unit for solving the above-described limitation will be described.

[0087] FIG. **9** is a cross-sectional view of a supporting unit according to an embodiment. Referring to FIG. **9**, the supporting unit may include a pair of support plates **350** in which the bar **31** and the support plate **35** are provided as one body. That is, the supporting unit may be provided by the same pair of support plates **350** rotated at a proper angle and coupled to each other in mirror symmetry. For this, the support plate **350** is provided symmetrically in FIG. **9**. Thus, the supporting unit may be assembled by coupling the pair of support plates **350** to face each other. Thus, the part may be shared, and stock costs may be reduced. A height of the radiation resistance sheet **32** may be supported by inserting the radiation resistance sheet **32** into a center while coupling the pair of support plates **350** to each other.

[0088] In the drawing, A and B indicate different types of bars. FIG. **10** is a view for explaining an A type bar. Referring to FIG. **10**, an A type bar may have a structure in which a high (elongated) column **352** provided on base **351** of either one (first) of the support plates **350** is coupled to a support groove **353** provided on the base **351** of the other (second) support plate **350**. As a height **H1** of the support groove **353** is low even if the high column **352** is large, the bar **31** is not broken even if dimensions are slightly wrong, or twisting occurs during fitting.

[0089] The radiation resistance sheet is not mounted to or does not contact the A type bar. A hole through which the high column **352** passes without contact may be provided in the radiation resistance sheet **32**.

[0090] The column includes a circular cross-section and may be manufactured in various shapes. In this specification, a diameter may mean a maximum distance connecting an outer circumference of the column.

[0091] FIG. **11** is a view for explaining a B-11 type bar. Referring to FIG. **11**, a B type bar may have a structure in which a multiple high (elongated) column **361** provided on the base **351** of either one (first) of the support plates **350** is inserted into an accommodation part or portion **365** provided on the base **351** of the other (second) support plate **350**.

[0092] It is seen that the B type bar is provided with B-11 and B-12 type bars provided in a vertically symmetrical manner. The first reference symbol B refers to a bar of the form constituted by the multiple high column and the accommodation part. The reference numeral **1** after the second dash (-) indicates that a length of multiple high column and accommodation part in the same B type. The first and second reference numerals **1** and **2** after the third dash (-) refer to the opposite in the up and down (vertical) direction. The same goes for the following.

[0093] Thus, FIG. **11** illustrates a B-11 type bar, and FIG. **12** illustrates a B-12 type bar. It is seen that the first and second reference numerals **1** and **2** after the third dash (-) are provided opposite to each other when viewed from a center of the support plate. This means that the same support plate is provided in a symmetrical structure.

[0094] The B type bar will be described in more detail. First, multiple high column **361** may include a large diameter portion **362** close to the base **351** and a small diameter portion **363** far from the base **351**. An upper end of the large diameter portion **362** forms a support protrusion **364**, and the radiation resistance sheet **32** may be seated thereon. An inclined insertion groove **366** is provided in the accommodation part **365** corresponding to the multiple high column **361** so that the smaller diameter portion **363** may be inserted without being damaged so that the diameter becomes larger toward an inlet thereof.

[0095] The accommodation part **365** may correspond to the support groove **353** provided in the A type bar and may be referred to as a support groove. In this case, it is understood that a height H.sub.2 of the support groove **365** provided for the B type bar is higher than a height H1 of the support groove **353** provided for the A type bar. In order that the radiation resistance sheet **32** is supported on the support protrusion **364**, and the insertion of the radiation resistance sheet **32** onto the small diameter portion **363** is conveniently performed, it is possible to design such that a diameter of the large diameter portion **362** is largest, a diameter of the small diameter portion **363** is smaller next (see reference numeral **322** in FIG. **13**) than the hole in the radiation resistance sheet, and the smallest diameter portion **363** has the smallest diameter.

[0096] Referring again to FIG. **9**, the B type bar may guide a position of the radiation resistance sheet **32**. The position of the radiation resistance sheet refers not only to movement of the radiation resistance sheet **32** in a two-dimensional planar direction, but also to a height thereof. That is, movement of the radiation resistance sheet **32** in the two-dimensional planar direction may be performed by a supporting action of the hole in the radiation resistance sheet **32** and the multiple high column **361**. A height of the radiation resistance sheet **32** may be supported at an interval between the support protrusion **32** and an end of the accommodation part **365**. Thus, there is no need to provide a space maintenance member that is inserted separately to maintain the space of the radiation resistance sheets.

[0097] As the B-12 type bar and the B-11 type bar are provided together, the height of the radiation resistance sheet **32** may be supported by only the support protrusion **32**. However, the height of the support protrusion **32** is provided to some extent to support the accurate height of the radiation resistance sheet **32**.

[0098] In the B type bar, the height H2 of the accommodation part **365** is higher than the height H1 of the support groove. The accommodation part **365** may not be provided in the same shape as a mold due to the resin material having low moldability as the height of the accommodation part **365** increases. For this, the number of B type bars may be smaller than the number of A type bars.

[0099] It is recommended that the number of B type bars is minimized to prevent damage to the bar as much as possible. However, it is desirable to have a minimum number of B type bars required for fixing the position of the radiation resistance sheet.

[0100] According to the supporting unit shown in FIG. **9**, the holes of the radiation resistance sheet **32** are aligned and inserted into the support protrusions **364** of the multiple high columns **361** of one of the support plates **350** disposed on a bottom side. The plurality of holes and the support protrusions **364** may be aligned so that positioning of the radiation resistance sheet is easily performed. Thereafter, the other (first) support plate **350** is aligned with the other (second) support plate **35** so as to be mirror-symmetrical with respect to the other support plate **35**, so that coupling of the supporting unit **30** is completed. The bar is not damaged even when coupling of the supporting unit **30** is completely performed in the vacuum evacuation process.

[0101] FIG. **13** is a view for explaining a coupling process of the supporting unit. Referring to FIG. **13**, it is seen that the multiple high columns **361** which are highlighted by a black circle and which provide the B type bar to the support plate **350** are provided symmetrically with respect to an oblique line. In addition, it is seen that the number is smaller than that of the A type bar. The radiation resistance sheet **32** is provided with hole **322** having a small size, which is capable of being caught by the support protrusion **364** of the multiple high column **361** and hole **321** having a

large size, which does not contact the high column but passes therethrough.

[0102] FIG. **14** is a cross-sectional view of a supporting unit according to another embodiment. Referring to FIG. **14**, there are two kinds of B type bars. The B type bars includes a B-21 type bar in which large diameter portion **362** of multiple high columns **361** has a height less than a height of small diameter portion **363** and a B31 type bar in which the large diameter portion **362** of the multiple high columns **361** has a height greater than a height of the small diameter portion **363**. The B-22 and B-32 type bars differ only in the position of the base **351** on which the multiple high columns **361** are formed. Of course, it is clearly understood from the drawings that the heights of the support protrusion **364** and the accommodation part **365** vary corresponding to the types of the multiple high columns, that is, B21 and B31 type bars.

[0103] In the present embodiment as well, the pair of support plates **350** is mirror-symmetrically provided to each other, so that the supporting unit **30** is provided by coupling the same pair of support plates **350** to the former embodiment. In the case of the present embodiment, two different radiation resistance sheets **32** may be provided. Thus, it may be clearly understood that radiation resistance is reduced. In the case of the present embodiment, the number of A type bars may be reduced to smoothly support the radiation resistance sheet **32**.

[0104] FIG. **15** is a cross-sectional view of a supporting unit according to further another embodiment. Referring to FIG. **15**, unlike the embodiment described above and the case in which the radiation resistance sheet **32** is supported at the interval between the support protrusion **364** of the column and the end of the accommodation part **365**, a feature in which it is supported only at the support protrusion of the support column end may be characteristically different.

[0105] FIG. **16** is a view for explaining a C type bar. Referring to FIG. **16**, a high (elongated) column **352** extending from the base **351** of one (first) side is supported through a method in which the support column **371** extending from the base **351** of the other (second) side is inserted. An end of the support column **371** forms a support protrusion **372**, and a position of the radiation resistance sheet **32** may be supported by the support protrusion **372** on a two dimensional plane.

[0106] The height of the radiation resistance sheet **32** may be held by the support protrusion **372** of a same kind but provided upside down. For example, the height of the radiation resistance sheet **32** may be maintained because the radiation resistance sheet **32** is supported in directions opposite to each other by the C-22 type bar and the C-22 type bar. It can be understood with reference to FIG. **15**.

[0107] There is a concern that the support column **371** may not be provided with a design shape due to a lack of moldability of the resin material. A different C type bar for solving this limitation is proposed.

[0108] FIG. **17** is a view for explaining a different C type bar. Referring to FIG. **17**, a middle column **375** extending from the base **351** of one (first) side is supported by a high support groove **376** extending from the base **351** of the other (second) side. An end of the high support groove **376** forms a support protrusion **377**, and the position of the radiation resistance sheet **32** may be supported by the support protrusion **377** on a two dimensional plane.

[0109] The height of the radiation resistance sheet **32** may be held by the support protrusion **377** of the same kind but provided upside down. For example, the height of the radiation resistance sheet **32** may be maintained because the radiation resistance sheet **32** is supported in directions opposite to each other by the C-11 type bar and the C-12 type bar. It can be understood with reference to FIG. **15**.

[0110] The C-11 and C-12 type bars show that the middle column **375** is slightly inserted into the high support groove **376**. Therefore, damage to the middle column **375** and the high support groove **376** due to insertion failure may be prevented.

[0111] The C-21 and C-22 type bars may be suitably used by the worker to manually fix the position of the pair of support plates. That is, when the worker attaches the C-21 type bar and the C-22 type bar between the plate members after assembling, the supporting unit may be completely

assembled in the vacuum evacuation process later.

[0112] In the case of the present embodiment, the A-type bar may also be provided. Also, in the present embodiment, it is possible to support two or more radiation resistance sheets by making the height of the support protrusions **377** and **372** various.

[0113] FIG. **18** illustrates graphs showing changes in adiabatic performance and changes in gas conductivity with respect to vacuum pressures by applying a simulation. Referring to FIG. **18**, it may be seen that, as the vacuum pressure is decreased, i.e., as the vacuum degree is increased, a heat load in the case of only the main body (Graph 1) or in the case where the main body and the door are joined together (Graph 2) is decreased as compared with that in the case of the typical product formed by foaming polyurethane, thereby improving adiabatic performance. However, it may be seen that a degree of improvement of the adiabatic performance is gradually lowered. Also, it may be seen that, as the vacuum pressure is decreased, gas conductivity (Graph 3) is decreased. However, it may be seen that, although the vacuum pressure is decreased, a ratio at which the adiabatic performance and the gas conductivity are improved is gradually lowered. Therefore, the vacuum pressure is decreased as low as possible. However, it takes a long time to obtain excessive vacuum pressure, and much cost is consumed due to excessive use of a getter. In the embodiment, an optimal vacuum pressure is proposed from the above-described point of view.

[0114] FIG. **19** is a graph illustrating results obtained by observing a time and a pressure in a process of exhausting the inside of the vacuum adiabatic body when a supporting unit is used. Referring to FIG. **19**, in order to create the vacuum space part **50** to be in the vacuum state, a gas in the vacuum space part **50** is exhausted by a vacuum pump while evaporating a latent gas remaining in parts of the vacuum space part **50** through baking. However, if the vacuum pressure reaches a certain level or more, there exists a point at which the level of the vacuum pressure is not increased any more ($\Delta t1$). After that, the getter is activated by disconnecting the vacuum space part **50** from the vacuum pump and applying heat to the vacuum space part **50** ($\Delta t2$). If the getter is activated, the pressure in the vacuum space part **50** is decreased for a certain period of time, but then normalized to maintain a vacuum pressure of a certain level. The vacuum pressure that maintains the certain level after activation of the getter is approximately 1.8×10^{-6} Torr.

[0115] In the embodiment, a point at which the vacuum pressure is not substantially decreased any more even though the gas is exhausted by operating the vacuum pump is set to a lowest limit of the vacuum pressure used in the vacuum adiabatic body, thereby setting a minimum internal pressure of the vacuum space part **50** to 1.8×10^{-6} Torr.

[0116] FIG. **20** is a graph obtained by comparing a vacuum pressure with gas conductivity. Referring to FIG. **20**, gas conductivities with respect to vacuum pressures depending on sizes of a gap in the vacuum space part **50** are represented as graphs of effective heat transfer coefficients (ek). Effective heat transfer coefficients (ek) were measured when the gap in the vacuum space part **50** has three sizes of 2.76 mm, 6.5 mm, and 12.5 mm. The gap in the vacuum space part **50** is defined as follows. When the radiation resistance sheet **32** exists inside of the vacuum space part **50**, the gap is a distance between the radiation resistance sheet **32** and the plate member adjacent thereto. When the radiation resistance sheet **32** does not exist inside of the vacuum space part **50**, the gap is a distance between the first and second plate members.

[0117] It was seen that, as the size of the gap is small at a point corresponding to a typical effective heat transfer coefficient of 0.0196 W/mK, which is provided to an adiabatic material formed by foaming polyurethane, the vacuum pressure is 2.65×10^{-1} Torr even when the size of the gap is 2.76 mm. It was seen that the point at which reduction in adiabatic effect caused by gas conduction heat is saturated even though the vacuum pressure is decreased is a point at which the vacuum pressure is approximately 4.5×10^{-3} Torr. The vacuum pressure of 4.5×10^{-3} Torr may be defined as the point at which the reduction in adiabatic effect caused by gas conduction heat is saturated. Also, when the effective heat transfer coefficient is 0.1 W/mK, the vacuum pressure is 1.2×10^{-2} Torr.

[0118] When the vacuum space part **50** is not provided with the supporting unit but provided with

the porous material, the size of the gap ranges from a few micrometers to a few hundreds of micrometers. In this case, the amount of radiation heat transfer is small due to the porous material even when the vacuum pressure is relatively high, i.e., when the vacuum degree is low. Therefore, an appropriate vacuum pump is used to adjust the vacuum pressure. The vacuum pressure appropriate to the corresponding vacuum pump is approximately 2.0×10^{-4} Torr. Also, the vacuum pressure at the point at which the reduction in adiabatic effect caused by gas conduction heat is saturated is approximately 4.7×10^{-2} Torr. Also, the pressure where the reduction in adiabatic effect caused by gas conduction heat reaches the typical effective heat transfer coefficient of 0.0196 W/mK is 730 Torr.

[0119] When the supporting unit and the porous material are provided together in the vacuum space part, a vacuum pressure may be created and used, which is in the middle between the vacuum pressure when only the supporting unit is used and the vacuum pressure when only the porous material is used.

[0120] In the description embodiments, a part for performing the same action in each embodiment of the vacuum adiabatic body may be applied to another embodiment by properly changing the shape or dimension of foregoing another embodiment. Accordingly, still another embodiment may be easily proposed. For example, in the detailed description, in the case of a vacuum adiabatic body suitable as a door-side vacuum adiabatic body, the vacuum adiabatic body may be applied as a main body-side vacuum adiabatic body by properly changing the shape and configuration of a vacuum adiabatic body.

[0121] The vacuum adiabatic body proposed in embodiments may be applied to refrigerators. However, the application of the vacuum adiabatic body is not limited to the refrigerators, and may be applied in various apparatuses, such as cryogenic refrigerating apparatuses, heating apparatuses, and ventilation apparatuses.

[0122] According to embodiments, the vacuum adiabatic body may be industrially applied to various adiabatic apparatuses. The adiabatic effect may be enhanced, so that it is possible to improve energy use efficiency and to increase the effective volume of an apparatus.

Claims

1. A vacuum adiabatic body comprising: a first plate; a second plate; a vacuum space provided between the first plate and the second plate and having a thickness in a first direction from the first plate toward the second plate; and a support configured to reduce deformation of the vacuum space and including a bar extending in the first direction, wherein the bar comprises a first portion and a second portion provided as a separate component from the first portion, to maintain a gap between the first plate and the second plate.
2. The vacuum adiabatic body of claim 1, wherein the first portion has a shape that is different from a shape of the second portion and comprises a part fastened to the second portion.
3. The vacuum adiabatic body of claim 1, wherein the second portion provided in a state separated from the first portion is assembled by placing the first and second portions in the vacuum space after the first portion and the second portion are fastened.
4. The vacuum adiabatic body of claim 1, wherein the second portion provided in a state separated from the first portion is assembled by placing the first and second portions in a space between the first and second plates after the first portion and the second portion are fastened.
5. The vacuum adiabatic body of claim 1, wherein the second portion provided in a state separated from the first portion is assembled by placing the first and second portions in a side of the first plate after the first portion and the second portion are fastened.
6. The vacuum adiabatic body of claim 1, wherein the first portion has a shape that is different from a shape of the second portion and comprises a part inserted into the second portion.
7. The vacuum adiabatic body of claim 1, wherein the first portion comprises a column.

- 8.** The vacuum adiabatic body of claim 1, wherein the second portion comprises a support groove in which the first portion is supported.
 - 9.** The vacuum adiabatic body of claim 1, wherein the second portion comprises an accommodation part in which the first portion is supported.
 - 10.** The vacuum adiabatic body of claim 1, wherein the second portion comprises a support column in which the first portion is supported.
 - 11.** The vacuum adiabatic body of claim 1, wherein the support further comprises a support plate extending in a second direction different from the first direction.
 - 12.** The vacuum adiabatic body of claim 1, wherein the support plate is disposed at an end of the bar, or is disposed to connect at least two bars to each other.
 - 13.** The vacuum adiabatic body of claim 1, wherein the support further comprises a porous material that is filled in the vacuum space and is made of material having a higher emissivity than material of the first and second plates.
 - 14.** A vacuum adiabatic body comprising: a first plate; a second plate; a vacuum space provided between the first plate and the second plate and having a width in a first direction from the first plate toward the second plate; and a support configured to reduce deformation of the vacuum space and including a bar extending in the first direction, wherein the bar comprises a column including an outer circumferential surface having a shape that varies along the first direction.
 - 15.** The vacuum adiabatic body of claim 14, wherein the bar comprises: a first portion including a first outer circumferential surface having a first size; a second portion including a second outer circumferential surface having a second size different from the first size; and a third portion to connect the first portion to the second portion.
 - 16.** A vacuum adiabatic body comprising: a first plate; a second plate; a vacuum space provided between the first plate and the second plate and having a width in a first direction from the first plate toward the second plate; and a support configured to reduce deformation of the vacuum space and including a bar extending in the first direction, wherein the bar comprises a first portion including a first column having a groove and a second portion including a second column inserted into the groove.
 - 17.** The vacuum adiabatic body of claim 16, wherein the first portion of the bar comprises: a first surface defining an inner circumferential surface formed by the groove; a second surface defining an outer circumferential surface; and a third surface to connect the inner circumferential surface to the outer circumferential surface.
 - 18.** An apparatus comprising a vacuum adiabatic body of claim 1.
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